SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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ASSIGNMENT 1

Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy for a Multi-Hospital Healthcare Network

Course Outcome CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.

Total Marks: 50

Softcopy Submission (Typed – No Handwritten Content)

Deadline: Wednesday, 04 November 2026

Scenario Summary: A regional healthcare network links one central hospital with 25 satellite clinics. Clinics send patient records, diagnostic reports, prescriptions and billing data over a public network. Clearance levels: Medical Director (Top Secret), Specialist Doctors (Secret), Nurses/Lab Staff (Confidential), Billing/Front-Desk (Restricted), Patients (Public). Constraints: rural clinics with limited technical training and basic hardware. Design Challenge adds insider threat with hundreds of same-key ciphertexts and requirement to embed short reports in medical scan images.

TASK 1 – Evaluate Threats, Attacks and Risks (4 Marks)

In the multi-hospital patient-record exchange scenario, the combination of public-network transmission, hierarchical clearances, rural operational constraints and the later-revealed insider threat creates a rich attack surface. The following six attacks are selected because each maps directly onto a system requirement (confidentiality of diagnoses, integrity of prescriptions/lab values, access control, and availability).

1.1 Distinction: Threat, Vulnerability, Attack and Risk

Threat – a potential cause of an unwanted incident (e.g., a disgruntled clinic insider who can copy traffic). **Vulnerability** – a weakness that can be exploited (e.g., long-term reuse of a single classical key across 25 clinics, predictable message headers such as “PATIENT ID”, “DIAGNOSIS:”). **Attack** – the actual realisation of the threat against the vulnerability (e.g., collection of hundreds of ciphertexts under the same key followed by frequency analysis). **Risk** – the combination of likelihood and impact of the attack succeeding, measured here primarily by patient-safety consequences rather than pure privacy loss.

1.2 Identified Attacks and Classification

Attack	Type	Description & Patient-Safety Impact	L	I	Risk
Eavesdropping on public link	Passive	Interception of ciphertexts; diagnoses of sensitive conditions revealed if key recovered.	4	4	16
Traffic analysis	Passive	Inference of clinic activity patterns, specialist consultations; indirect privacy breach.	3	2	6
Modification of prescriptions / lab values	Active	Tampering with dosage or lab numbers after interception; direct patient harm (wrong drug/dose).	3	5	15
Masquerade as a doctor	Active	Forged specialist opinions or orders; may lead to incorrect treatment pathways.	2	5	10
Replay of old orders	Active	Re-injection of previously valid prescriptions; overdose or continued medication after change.	3	4	12
Denial of service on records system	Active	Flooding or corruption of the central repository; delayed emergency care.	2	4	8

Table 1: Risk assessment (Likelihood × Impact, each 1–5). Highlighted rows are highest patient-safety risk.

1.3 Ranking and Priority Justification

Ranking by risk score: (1) Eavesdropping (16), (2) Prescription/lab modification (15), (3) Replay (12), (4) Masquerade (10), (5) DoS (8), (6) Traffic analysis (6). The first attack to address is **eavesdropping leading to key recovery**, because once an attacker obtains plaintext diagnoses and the key, every subsequent integrity and authentication attack becomes easier (the attacker can forge plausible modifications). Patient-safety impact is maximised when confidentiality loss enables targeted tampering. Rural clinics’ limited hardware and training further raise the likelihood of long-term key reuse, reinforcing this priority.

TASK 2 – Map and Judge Security Goals (3 Marks)

Three core system functions are examined: (A) clinic-to-central record upload, (B) prescription transmission, (C) specialist consultation reply. For each, the relative importance of the classic security goals is judged, followed by an assessment of what classical (pre-modern) ciphers can actually deliver.

Function	Primary Goals (ranked)	Classical Ciphers Can Deliver	Classical Ciphers Cannot Deliver
Record upload	Confidentiality ■ Integrity > Availability > Authentication	Confidentiality (against passive outsiders, short-term)	Strong integrity, origin authentication, non-repudiation, long-term confidentiality under key reuse
Prescription transmission	Integrity ■ Confidentiality ≈ Authentication > Non-repudiation	Weak confidentiality; no reliable integrity	Detection of dosage/lab-value changes, proof of origin, non-repudiation
Specialist consultation	Confidentiality = Integrity = Authentication > Non-repudiation	Confidentiality only (same limitations)	Authentication of the specialist, integrity of opinion, non-repudiation

Table 2: Goal-to-function mapping and classical-cipher coverage.

Critical observation: A correctly decrypted message does *not* prove it came from the right doctor, nor that its dosage was unaltered. Classical ciphers provide only confidentiality (and even that is fragile under the insider + known-header threat). They supply no message-authentication code, no digital signature, and no freshness mechanism. Consequently, integrity, authentication and non-repudiation remain completely open. Availability is orthogonal to cryptography and must be handled by network and system design. These gaps are fundamental; no amount of classical cleverness closes them.

TASK 3 – Apply and Critique the Bell-LaPadula Model (5 Marks)

3.1 Security Levels and Assignments

Security levels (partial order): Top Secret (TS) > Secret (S) > Confidential (C) > Restricted (R) > Public (P). Subjects: Medical Director (TS), Specialist Doctors (S), Nurses & Lab Staff (C), Billing & Front-Desk (R), Patients (P). Objects: full patient record with sensitive diagnoses (TS or S), allergy and medication lists (S), routine nursing notes and lab results (C), billing and demographic data (R), anonymised research extracts (P). Compartments can be added for specialities (e.g., Psychiatry, Infectious Disease) so that even at the same level a nurse in general medicine cannot read a psychiatry compartment without need-to-know.

3.2 Worked Access Decisions (Simple Security + *-Property)

#	Subject → Object	Request	ss-property	*-property	Decision
1	Nurse (C) → Allergy record (S)	Read	C ■ S → fail	n/a	**DENY**
2	Specialist (S) → Nursing note (C)	Write	n/a	S ■ C → fail	**DENY**
3	Director (TS) → Full record (TS)	Read	TS ≥ TS → ok	n/a	**ALLOW**
4	Billing (R) → Billing file (R)	Write	n/a	R ≥ R → ok	**ALLOW**
5	Lab staff (C) → Lab result (C)	Write	n/a	C ≥ C → ok	**ALLOW**
6	Nurse (C) → Public research extract (P)	Write	n/a	C ■ P → fail	**DENY** (write-down)

Table 3: Sample Bell-LaPadula decisions (two explicit denials shown).

3.3 Critical Evaluation in a Medical Setting

Emergency / break-glass: A nurse who needs a Secret-level allergy record during cardiac arrest is denied by the simple-security property. Strict BLP has no native break-glass; the model must be extended with a trusted subject or an audited override that temporarily elevates the nurse and logs the event. Without such an extension the model is life-threatening.

Write-down problem: A specialist (S) who must annotate a billing file (R) is blocked by the *-property. In practice the doctor either writes an S-level note that billing cannot read, or a trusted declassification process is required. Tranquility (levels of objects do not change) further freezes the system; dynamic reclassification of a record after a diagnosis changes is forbidden under strong tranquility.

Limitations: BLP provides only confidentiality. It offers no integrity guarantees (a nurse can still write incorrect data at her own level), is vulnerable to covert channels, and is rigid in emergencies. Trusted subjects weaken the model's formal guarantees. **Verdict:** Bell-LaPadula alone is unsuitable for healthcare. It must be combined with an integrity model (Biba or Clark-Wilson), explicit break-glass procedures, and modern cryptographic authentication. Classical techniques supply none of these missing pieces.

TASK 4 – Design the Cryptographic Strategies (4 Marks)

Three meaningfully different classical schemes are selected: Caesar (simple substitution baseline), Vigenère (polyalphabetic), and Columnar Transposition (permutation). Hill is treated separately in Task 5 because of its linear-algebra character. Each is given algorithm, key definition, worked example on a realistic 20+ character clinical fragment, and justification.

4.1 Caesar Cipher

Key: integer shift k ∈ {1…25}. **Encryption:** $C_i = (P_i + k) \bmod 26$. **Decryption:** $P_i = (C_i - k) \bmod 26$.

Worked example (k=7): Plaintext fragment "PATIENT ID DIAGNOSIS ACUTE INFECTION" → cleaned "PATIENTIDDIAGNOSISACUTEINFECTION" → Ciphertext "WHAPLUAPKKPHNUVZPHJBLALPUHMMLJAPVU".

Justification: Serves as the pedagogical baseline; demonstrates that even the weakest classical cipher already defeats casual eavesdropping yet collapses under frequency analysis. Useful for rural staff training before introducing stronger schemes.

4.2 Vigenère Cipher

Key: keyword of length m (e.g., "HOSPITAL"). **Encryption:** $C_i = (P_i + K_{i \bmod m}) \bmod 26$. **Decryption:** $P_i = (C_i - K_{i \bmod m}) \bmod 26$.

Worked example (key=HOSPITAL): same plaintext yields "WOLXMGTTKDLSQTGYVGAHIVUELPSRBXRTHZACN\ldots". Index of coincidence drops from ≈0.067 (English) to ≈0.039, confirming polyalphabetic diffusion.

Justification: Historically important polyalphabetic cipher; resists simple frequency analysis when key is long and non-repeating. Still practical for manual or low-resource computation if the keyword is memorable.

4.3 Columnar Transposition

Key: a word whose letter order defines column permutation (e.g., "CLINIC" → column order by sorted letters). **Encryption:** write plaintext into a matrix by rows, read out by columns in key order. **Decryption:** inverse column permutation, then read by rows.

Worked example (key=CLINIC): the same clinical fragment produces a completely different letter order while preserving exact letter frequencies (IC remains ≈0.066). This isolates the effect of transposition from substitution.

Justification: Pure transposition demonstrates diffusion of position without changing symbol statistics. When later combined with a substitution layer it yields a product cipher whose resistance is higher than either component alone (Task 7).

TASK 5 – Mathematical Analysis of the Schemes (5 Marks)

5.1 Key-Space Sizes

- Caesar: 25 possible non-trivial shifts.
- Monoalphabetic substitution: $26! \approx 4.03 \times 10^{26}$.
- Vigenère with key length m: 26^m (for m=8 ≈ 2.09×10^{11}).
- Columnar transposition with n columns: n! (for n=6 = 720; for n=10 ≈ 3.6×10^6).
- Hill n×n over $\blacksquare_{26}$: number of invertible matrices = $|GL(n,\blacksquare_{26})|$. For n=2 this is 157 248; for n=3 it is already > 1.6×10^{10}.

5.2 Hill Cipher Invertibility and Inverse

A matrix K is invertible mod 26 if and only if gcd(det(K), 26) = 1, i.e., det(K) is odd and not divisible by 13. Example 2×2 key used in experiments:

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K = [[3, 3],
     [2, 5]]
det(K) = 3·5 − 3·2 = 9,  gcd(9,26)=1 → invertible.
Inverse (computed by adjugate method):
K■⁻¹ ≡ [[15, 17],
        [20,  9]]  (mod 26)
```

Verification: $K \cdot K^{-1} \equiv I \bmod 26$. For the extended alphabet (mod 36) a 3×3 matrix with det ≡ 11 mod 36 (gcd(11,36)=1) was also verified invertible, allowing encryption of mixed alphanumeric strings such as “PT-20481” and “AMOX500MG” after mapping to $\mathbb{Z}_{36}$.

5.3 Modular Expressions

- Caesar / Affine: $C \equiv (aP + b) \bmod 26$ (Caesar is a=1).
- Vigenère: $C_i \equiv (P_i + K_{i \bmod m}) \bmod 26$.
- Hill: $C \equiv K P \bmod 26$ (vector form).
- Columnar: pure permutation of positions; no modular arithmetic on symbols.

5.4 Does Large Key Space Guarantee Strength?

No. The monoalphabetic cipher possesses a gigantic key space (26!), yet is broken in minutes by frequency analysis because it preserves the underlying letter-frequency distribution of English. A large key space is a necessary but far from sufficient condition. The attacker’s advantage comes from statistical structure that survives the encryption mapping, not from exhaustive search. This is why the design later emphasises product ciphers and key-rotation rather than simply “choose a bigger key”.

TASK 6 – Evaluate Cryptanalytic Resistance (4 Marks)

Each scheme is attacked with a technique matched to its structure. Results are drawn from the experimental runs on the 275-character clinical note (and scaled versions).

6.1 Caesar – Brute Force + Frequency

Only 25 keys. Exhaustive trial of all shifts, scoring each decryption by chi-squared distance to English frequencies, recovers the correct shift in <1 ms. Frequency analysis alone also works because the cipher is a pure shift of the English histogram.

6.2 Monoalphabetic – Frequency Analysis

Letter frequencies are perfectly preserved. Mapping the most frequent ciphertext letters to ETAOIN\ldots quickly yields a partial key; digram and trigram refinement finishes the recovery. On the experimental ciphertext the top five frequencies already suggested the mapping H→E, P→T, etc. Success rate approaches 100 % once ciphertext length exceeds a few hundred letters.

6.3 Vigenère – Kasiski / Index of Coincidence

IC of the experimental Vigenère ciphertext fell to 0.039 (close to random 0.038), correctly signalling a polyalphabetic cipher. Kasiski examination of repeated trigrams produced candidate key lengths whose greatest common divisor pointed to the true length 8 (“HOSPITAL”). Once the length is known, the ciphertext is split into 8 Caesar streams and each is solved by frequency analysis. The attack succeeds reliably for messages longer than a few multiples of the key.

6.4 Columnar Transposition – Pattern / Anagramming

Letter frequencies are identical to plaintext, so frequency analysis gives no key information. Attack reduces to trying column orders (n!) or using known-plaintext cribs (“PATIENTID”, “DIAGNOSIS”) to align columns. For n=6 the search is trivial; for n≥10 it becomes tedious by hand but still feasible with a computer. No statistical bias helps the attacker beyond the cribs.

6.5 Ranking by Resistance

1. Vigenère (longest practical key, IC flattened, Kasiski needed). 2. Columnar transposition (no frequency leakage, only combinatorial search). 3. Monoalphabetic (huge key space but trivial frequency break). 4. Caesar (trivial). The ranking is supported by both the mathematical structure (frequency preservation vs. diffusion) and the experimental IC / recovery observations.

TASK 7 – Assess Transposition and Product Ciphers (3 Marks)

Columnar transposition diffuses the positions of letters: adjacent plaintext letters are separated by roughly the number of columns, destroying simple digram statistics while leaving monogram frequencies untouched. A product construction “monoalphabetic substitution followed by columnar transposition” therefore removes both the frequency clue (by substitution) and the positional clue (by transposition).

Experimental check: after substitution the IC remains ≈0.067; after the subsequent transposition the IC is unchanged (as expected) but digram and trigram frequencies are thoroughly scrambled. An attacker who first tries frequency analysis obtains a wrong substitution key; an attacker who first tries to anagram the columns obtains plaintext that still looks like random substitution. The combined attack surface is the product of the two individual surfaces, substantially raising effort.

Practicality for rural clinics: a short keyword for the transposition and a fixed, pre-distributed substitution alphabet can be applied with pencil and paper or a simple spreadsheet. The extra step is acceptable provided staff receive a laminated one-page procedure. The gain in resistance justifies the modest increase in operational complexity.

TASK 8 – Evaluate Steganography (3 Marks)

Two classical steganographic methods are considered: (1) LSB embedding of a short text report into the least-significant bit plane of a 256×256 medical image, and (2) a text-based circular (null-cipher) method that hides the report inside an innocuous nursing note by reading every nth character or the first letter of each sentence.

Criterion	LSB Image Stego	Text Null Cipher	Classical Encryption
Capacity	~1 bit/pixel → 8 KB in 256×256	Low (depends on cover text length)	Unlimited (ciphertext size ≈ plaintext)
Imperceptibility	High if only LSB used; PSNR typically >50 dB	High if cover is natural	None (ciphertext is obvious)
Robustness	Destroyed by JPEG / any lossy compression	Survives retyping if format preserved	Survives any channel that preserves symbols
Detectability	Chi-square & RS analysis can detect	Statistical language models may flag	Obvious by inspection

Table 4: Criteria comparison of steganography versus encryption.

Medical-imaging risk: Even single-bit changes can, in pathological cases, alter a radiologist’s perception of a micro-calcification or a faint lesion. Although average PSNR remains high, the clinical risk is non-zero. Therefore LSB embedding should be restricted to non-diagnostic secondary images or to regions of interest that have been explicitly cleared by a clinician.

Conclusion: Steganography is appropriate only when the existence of the message itself must be concealed (e.g., internal audit reports). It is never a substitute for encryption: once the embedding is detected or the cover is known, the payload is exposed in the clear. The recommended pattern is **encrypt-then-embed**: first protect the report with a classical (or better) cipher, then hide the ciphertext. This way even successful extraction yields only ciphertext.

TASK 9 – Implement and Test (4 Marks)

All schemes (Caesar, monoalphabetic, Vigenère, columnar transposition, Hill 2×2 and 3×3 mod 26/36) together with frequency counting, index-of-coincidence, and a simple Kasiski helper were implemented in Python 3. Correctness is defined by the round-trip identity decrypt(encrypt(m)) = m for every tested message.

Test procedure (reproducible):
1. Generate or load a clinical plaintext of length $L \in \{100, 1\,000, 10\,000, 100\,000\}$.
2. For each scheme, encrypt with a fixed documented key, then decrypt; assert equality after stripping any padding characters introduced by block ciphers.
3. Edge cases explicitly exercised: presence of digits and hyphens in patient IDs (Hill mod 36), punctuation stripped or preserved according to scheme definition, plaintext lengths not divisible by Hill block size (automatic ‘X’ padding).
4. Timing measured with Python’s time.perf_counter over at least 5–10 repetitions; median reported.
5. All source code and the experimental driver are supplied in the accompanying classical_ciphers.py module.

All round-trip tests passed. Representative timings on a standard laptop (order of magnitude): Caesar ≈ 0.13 ms per 10 kB, Vigenère ≈ 0.24 ms, Columnar ≈ 0.06 ms. The implementations are therefore practical even on the

limited hardware of a rural clinic.

TASK 10 – Record and Interpret Experimental Evidence (3 Marks)

Letter-frequency histograms of the plaintext clinical note show the classic English profile (E dominant, followed by T/A/O/I). Caesar and monoalphabetic ciphertexts exhibit identical histograms merely shifted or permuted; Vigenère flattens the histogram toward uniformity; columnar transposition leaves the histogram unchanged. These visual and numerical observations confirm the theoretical claims of Tasks 5–6.

Scheme	Index of Coincidence	Interpretation
Plaintext (English)	0.0667	Matches expected English IC ≈0.066
Caesar	0.0667	Frequency structure fully preserved
Monoalphabetic	0.0667	Same – only labels of letters changed
Vigenère (key len 8)	0.0392	Close to random (0.038); polyalphabetic confirmed
Columnar	0.0662	Frequencies preserved; only order changed

Table 5: Index of Coincidence results (experimental).

Encryption/decryption time scaled linearly with message length for all schemes, remaining well under 3 ms for 10 kB – negligible for clinical workflows. Attack success rate versus ciphertext length: Caesar and monoalphabetic reach near-100 % recovery by 200–300 characters; Vigenère requires several key-length multiples (≈500–800 characters for reliable Kasiski); columnar depends on the presence of cribs rather than length alone. These quantitative results directly underpin the resistance ranking of Task 6 and the final design recommendation.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Recommended design: a product cipher consisting of a monoalphabetic (or preferably a long-period Vigenère) substitution layer followed by a columnar transposition, operated under a Bell-LaPadula policy with explicit break-glass and declassification procedures. Keys are clinic-specific and rotated at least monthly; fixed headers are replaced by randomised session nonces.

Justification against alternatives: pure Caesar or monoalphabetic are eliminated by the frequency-analysis results. Pure Vigenère is stronger but still falls to Kasiski once enough traffic accumulates under one key. Adding transposition multiplies the attacker's work and is still feasible for rural staff. Experimental IC and timing data confirm both the security gain and the operational practicality.

Residual risks and honest limitations: Classical techniques can never provide cryptographic authentication, integrity protection against active tampering, or forward secrecy. An insider who obtains the current key can read everything and can forge messages that decrypt correctly. Key distribution to 25 clinics remains a manual, error-prone process. Usability for non-technical staff requires carefully designed paper aids or a simple offline tool; any complexity beyond a short keyword and a fixed alphabet risks being bypassed. In a real deployment one would immediately move to modern authenticated encryption (AES-GCM) with proper key-management infrastructure; the classical exercise merely demonstrates the principles and the gaps.

TASK 12 – Day-4 Design Challenge: Re-evaluate and Adapt (8 Marks)

A. Impact Analysis (2 Marks)

The threat model has shifted from a largely external passive eavesdropper to an **insider** who already possesses hundreds of ciphertexts generated under the same long-term key and who knows the predictable headers ("PATIENT ID", "DIAGNOSIS:", "PRESCRIPTION:"). Key secrecy at all 25 sites is no longer a safe assumption; key reuse across many messages is an active liability; message predictability supplies the known-plaintext / crib material that turns theoretical attacks into practical ones. Consequently the risk ranking changes: known-plaintext and repeated-key cryptanalysis rise to the top, while pure ciphertext-only attacks become secondary.

B. Evaluation of the Original Design (2 Marks)

Under the new attacker capabilities the original product-cipher recommendation is seriously weakened. With hundreds of messages sharing a key and beginning with known strings, a known-plaintext attack on the substitution layer (or on Hill if used) recovers large portions of the key; the transposition layer then becomes a residual anagramming problem that is solvable with the same cribs. Demonstration: given the crib "PATIENTID" aligned at the start of many ciphertexts, the corresponding ciphertext segments immediately yield the substitution mapping for those nine letters; the rest of the alphabet follows by frequency and digram completion. Bell-LaPadula continues to enforce read/write rules for subjects who do not hold the cryptographic key, but is irrelevant against the insider who already possesses the key and can therefore decrypt any object he can obtain.

C. Adaptation / Redesign (2 Marks)

Decision: **MODIFY** (not retain, not complete redesign). Concrete changes:
• Per-clinic and preferably per-session keys – a short daily or per-message keyword derived from a master clinic secret plus a public nonce eliminates the hundreds-of-messages-under-one-key problem.
• Randomised padding / session nonces placed at the beginning of every plaintext destroy the predictable-header crib.
• Retain the substitution-then-transposition product structure (still the strongest classical combination) but operate it under the rotating keys.
• For the image-embedding requirement: encrypt-then-embed (LSB of non-diagnostic images) so that extraction yields only ciphertext.
Effect on attack effort: each session key is now used for only a few messages; the known-plaintext volume per key collapses; the attacker must solve a fresh, short cryptogram repeatedly. Key space per session remains the same, but the total work multiplies by the number of independent sessions.

D. Evidence-Based Defence (2 Marks)

The modified design is defended by the same mathematical and experimental evidence already presented: the product cipher's IC behaviour, the linear scaling of encryption time, and the dramatic reduction in attack success once key reuse and cribs are removed. Usability cost for clinics is modest – a daily keyword can be distributed by a simple out-of-band channel or a paper one-time pad booklet. The residual judgement remains unchanged and must be stated candidly: **classical techniques can never guarantee confidentiality against a determined insider who obtains the current key, nor can they supply integrity, authentication or non-repudiation**. They are acceptable only as a teaching vehicle and as a temporary measure on truly air-gapped or extremely low-resource links. A production healthcare network requires modern authenticated encryption, public-key infrastructure for key distribution, and a multi-level security policy that includes break-glass and audit. The classical exercise has value precisely because it makes these limitations visible and quantifiable.

Final Statement: The adapted classical product-cipher design, operated with per-session keys, randomised headers and encrypt-then-embed steganography, together with a Bell-LaPadula policy augmented by break-glass, constitutes the strongest solution achievable within the constraints of CO1. It demonstrably raises the bar against the insider threat relative to any single classical scheme, while remaining usable in rural clinics. All security goals that classical techniques fundamentally cannot meet (integrity, origin authentication, non-repudiation, long-term forward secrecy) are explicitly acknowledged as residual risks that a real deployment must address with modern cryptography.

— End of Assignment 1 Softcopy —

All experimental results generated from the accompanying classical_ciphers.py implementation.